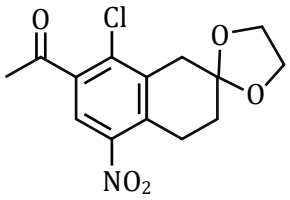
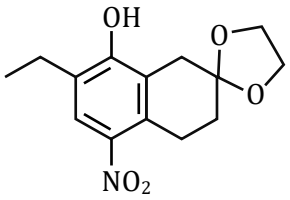
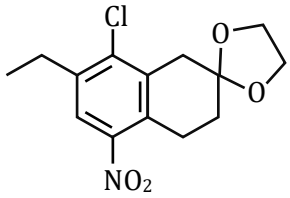
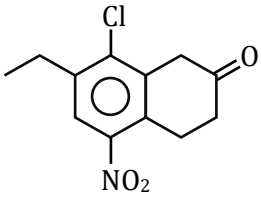
Carbonyl Compounds : RACE-01

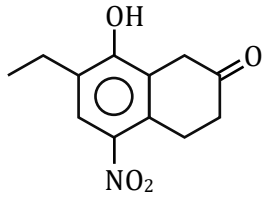
1. 

$\xrightarrow[\text{(Wolff-Kishner reduction)}]{N_2H_4/KOH/H_2O}$ (A) ; Product A is :
major

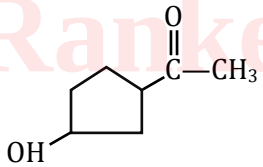
(A) 

(B) 

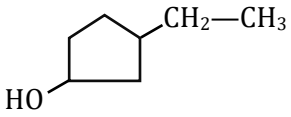
(C) 

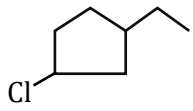
(D) 

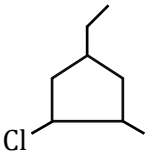
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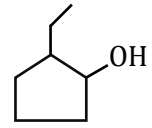
2. 

$\xrightarrow[HCl]{Zn(Hg)}$ (A) ; Identify the A .

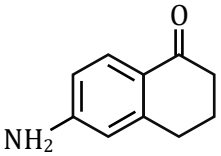
(A) 

(B) 

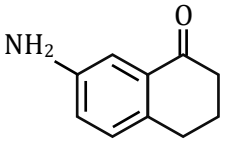
(C) 

(D) 

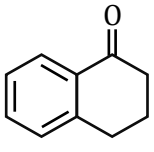
3. Increasing order of equilibrium constants for the formation of a hydrate:



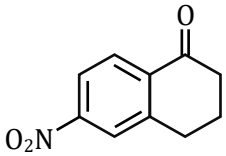
(I)



(II)



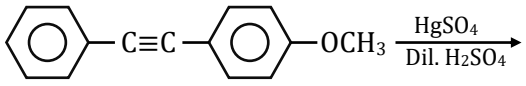
(III)



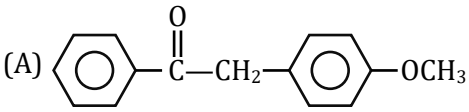
(IV)

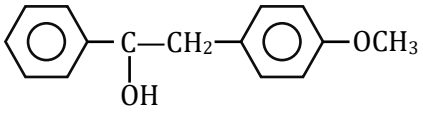
(A) IV < III < II < I
(B) IV < III < I < II
(C) I < II < III < IV
(D) II < III < I < IV

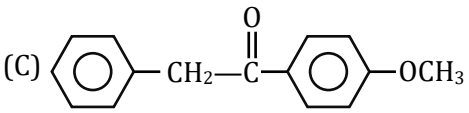
For More Join: @IITJEE_Advanced

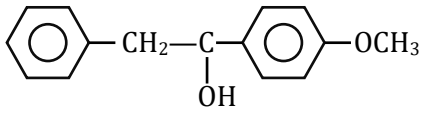
4. 

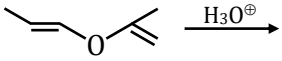
(A) Major product Product (A) is :

(A) 

(B) 

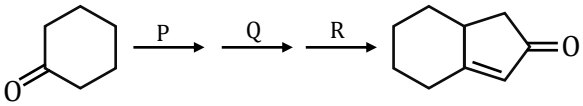
(C) 

(D) 

5. 

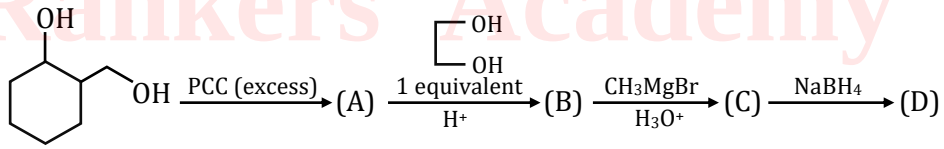
A + B.Compound (A) & (B) can be differentiated by :

(A) 2-4-DNP (B) Fehling solution (C) Lucas reagent (D) $NaHSO_3$

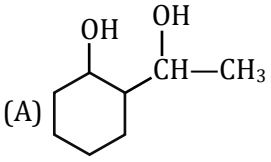
6. 

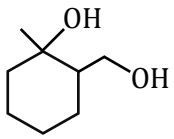
Reagents to carry out above conversion, P, Q, R respectively are :
(A) $H_2C{=}CH{-}CH_2{-}Br$, $(HO^{\ominus})$, $[HO^{\ominus}, \Delta]$, Wacker-process
(B) $H_2C{=}CH{-}CH_2{-}Br(HO^{\ominus})$, Wacker-process, $HO^{\ominus}\ \Delta$
(C) Wacker-process, $H_2C{=}CH{-}CH_2{-}Br(HO^{\ominus})$, $HO^{\ominus}(\Delta)$
(D) Wacker-process, $HO^{\ominus}(\Delta)$, $H_2C{=}CH{-}CH_2{-}Br(HO^{\ominus})$

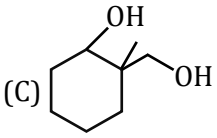
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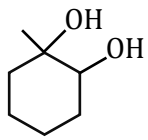
7. 

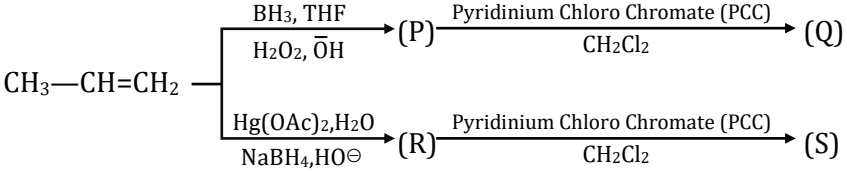
Product (D) will be:

(A) 

(B) 

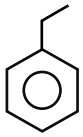
(C) 

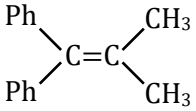
(D) 

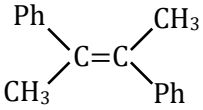
8. 

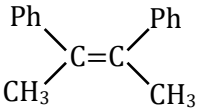
Relationship between products (Q) and (S) is:
(A) Positional isomer (B) Chain isomer
(C) Stereoisomer (D) Functional isomer

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9. $C_{16}H_{16}$ (A) $\xrightarrow{O_3}$ (B) C_8H_8O $\xrightarrow[HO^-/\Delta]{NH_2—NH_2}$  ; Reactant (A) in this reaction is:

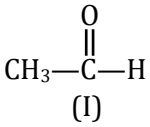
(A) 

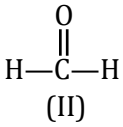
(B) 

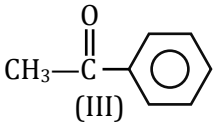
(C) 

(D) both (B) and (C)

10. Correct order of reactivity of following compounds towards Grignard reagent ?

 (I)

 (II)

 (III)

(A) I > II > III (B) II > I > III (C) II > III > I (D) I > III > II

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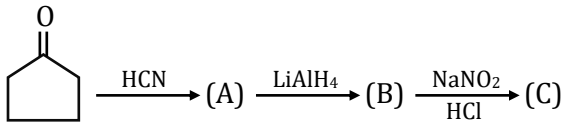
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	B	C	C	B	B	B	D	D	B

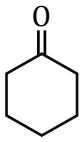
For More Join: @IITJEE_Advanced

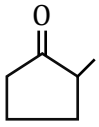
RACE
Regular Analysis Through Continuous Exercise

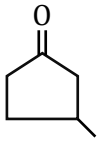
Carbonyl Compounds : RACE-02

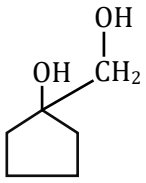
1. 

End product (C) in above reaction is :

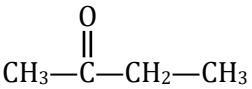
(A) 

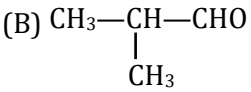
(B) 

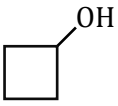
(C) 

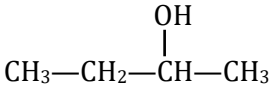
(D) 

2. Compound (X) C_4H_8O, which reacts with 2, 4-DNP derivative and gives negative haloform test is :

(A) 

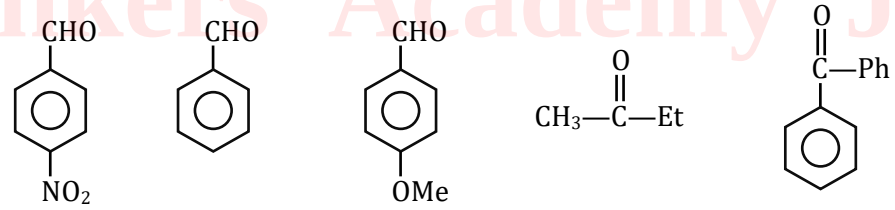
(B) 

(C) 

(D) 

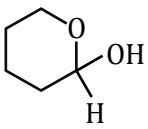
3. The correct order of rate of reaction toward nucleophilic addition reaction:

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(a) (b) (c) (d) (e)

(A) a > b > c > d > e (B) a > B > d > c > e (C) a > d > e > b > c (D) a > b > e > d > c

4. The structure 

would be best classified as a (an) :

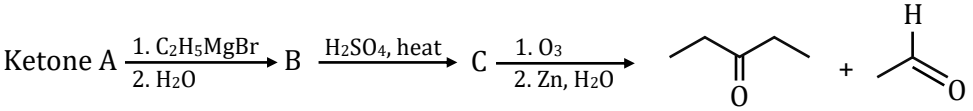
(A) Acetal (B) Hemiacetal (C) Hydrate (D) Cyanohydrin

5. The reaction of $C_6H_5CH{=}CHCHO$ with $LiAlH_4$ gives:

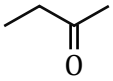
(A) $C_6H_5CH_2CH_2CH_2OH$ (B) $C_6H_5CH{=}CHCH_2OH$

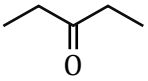
(C) $C_6H_5CH_2CH_2CHO$ (D) $C_6H_5CH_2CHOHCH_3$

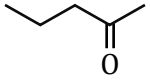
6. Consider the following sequence of reactions.



The ketone (A) is:

(A) 

(B) 

(C) 

(D) 

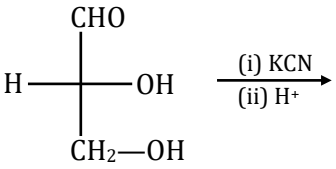
For More Join: @IITJEE_Advanced

7. The presence of unsaturation in organic compounds can be tested with :

(A) Schiff's reagent (B) Tollens' reagent (C) Fehling's reagent (D) Baeyer's reagent

8. Which of the following gives iodoform test ?

(A) CH_3CH_2OH (B) C_2H_5CHO (C) $(CH_2OH)_2$ (D) None of these

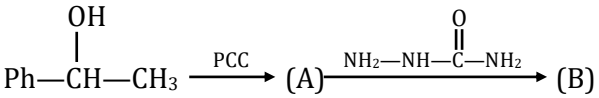
9. 

Product ; Product obtained in the reaction is :

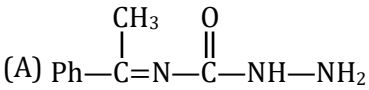
D-(+)-Glyceraldehyde

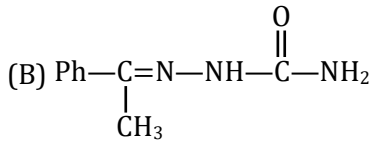
(A) Diastereomer (B) Racemic

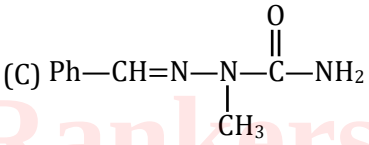
(C) Meso (D) Optically Pure enantiomer

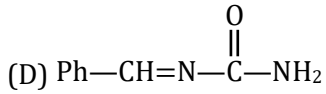
10. 

Product (B) is:

(A) 

(B) 

(C) 

(D) 

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ANSWER KEY

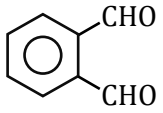
Question	1	2	3	4	5	6	7	8	9	10
Answer	A	B	A	B	A	B	D	A	A	B

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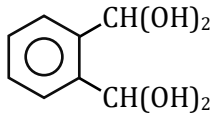
RACE
Regular Analysis Through Continuous Exercise

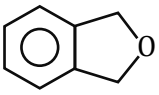
Carbonyl Compounds : RACE-03

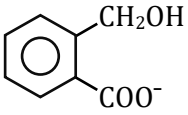
1. The reaction,

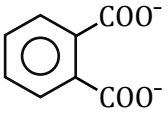


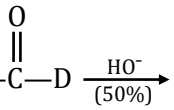
+ conc. NaOH $\xrightarrow{\text{heat}}$, Products Identify the product.

(A) 

(B) 

(C) 

(D) 

2. 

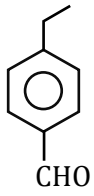
; Product of this Cannizaro reaction is:

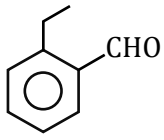
(A) $D—CO_2^-$ + CH_2DOD (B) $H—CO_2^-$ + $D—CO_2^-$

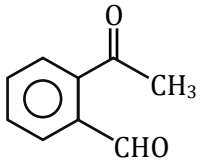
(C) $D—CO_2^-$ + CH_2DOH (D) $D—CO_2^-$ + CHD_2OH

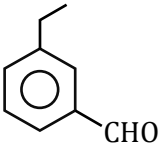
3. An organic compound with the molecular formula $C_9H_{10}O$ forms a 2, 4-DNP derivative, reduces Tollen's reagent and undergoes Cannizaro reaction, on vigorous oxidation it gives 1, 2-benzenedicarboxylic acid. Structure of organic compound is:

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(A) 

(B) 

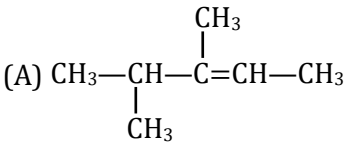
(C) 

(D) 

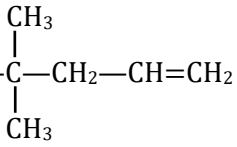
4. $(A)_{C_7H_{14}} \xrightarrow[Zn/AcOH]{O_3} (B) + (C)$

Compound (A) exist in geometrical isomers and (B) gives Cannizaro reaction.

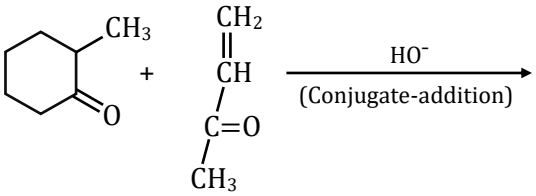
(A) will be:

(A) 

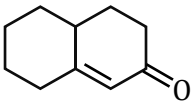
(B) $(CH_3)_3CCH_2—CH=CH_2$

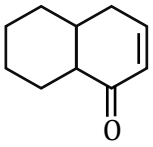
(C) $(CH_3)_3C—CH=CH—CH_3$ (D) 

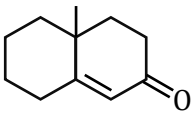
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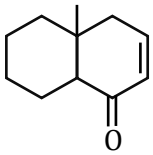
5. 

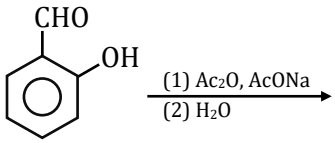
(A) ; Product (A) is :

(A) 

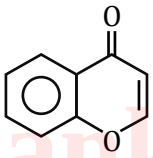
(B) 

(C) 

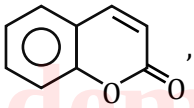
(D) 

6. 

(A); Identify product and name of reaction.

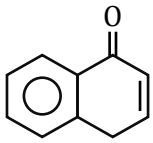
(A) 

, Aldol condensation

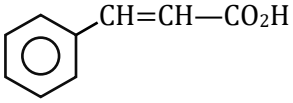
(B) 

, Perkin reaction

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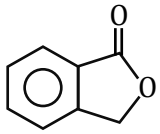
(C) 

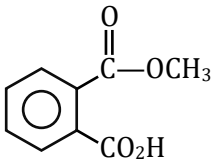
, Cannizaro reaction

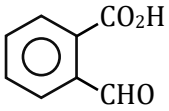
(D) 

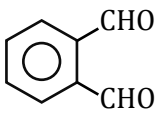
, Claisen-condensation

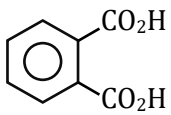
7. Which of the following reactant on reaction with conc. NaOH followed by acidification gives the following lactone as the product ?



(A) 

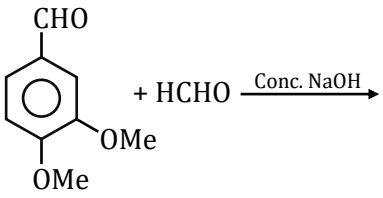
(B) 

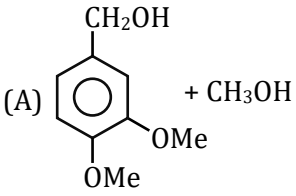
(C) 

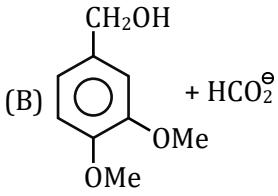
(D) 

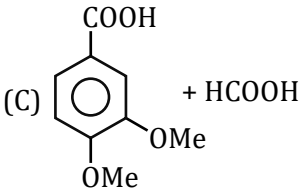
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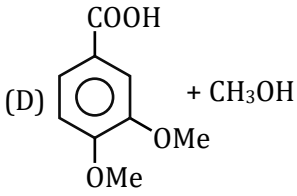
8. The following reaction gives :

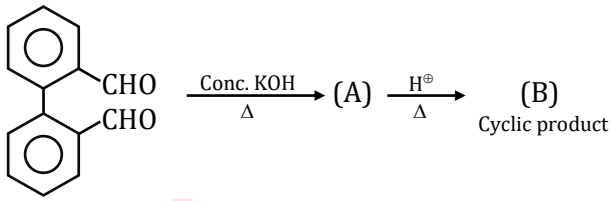


(A) 

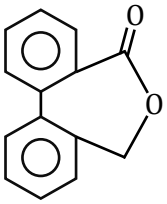
(B) 

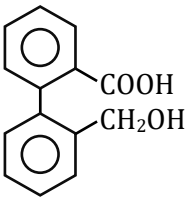
(C) 

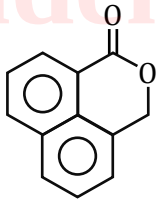
(D) 

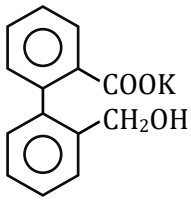
9. 

Structure of (B) is:

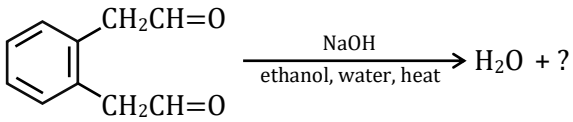
(A) 

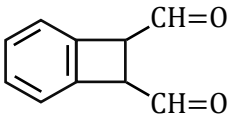
(B) 

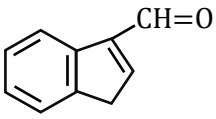
(C) 

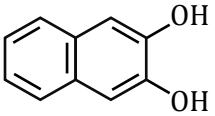
(D) 

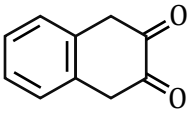
10. Identify the principal product of the following reaction?



(A) 

(B) 

(C) 

(D) 

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	B	C	C	B	C	B	A	B

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